

LINEAR AND ROBUST CONTROL SYSTEMS

ELEC 9731

2 - SISO System Analysis

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Topics

- 1** Review Definitions -system, model etc
- 2** Models and Tools: ODE, State Space;
Laplace, Fourier
- 3** State Space - ODE: Connexions
- 4** Modal Description
- 5** Controllability, Observability
- 6** Modal View of Controllability, Observability
and Hidden Modes

System

- System

A *system* is a medium that relates a cause to an effect or an input to an output. When the cause is assigned a value it becomes a quantity or variable. And a quantity that varies with time is a *signal*.

- Model

If the relation among the input/output signals is expressed mathematically we have a mathematical *model* of the system. *Parameters* are derived quantities expressing system properties e.g.

R, C, L . More generally parameters are *slowly* time varying internal signals.

System description can be determined :

(i) theoretically e.g. from the laws of physics
or (ii) empirically e.g. measure input/output signals and determine the equations relating them by statistical methods. This is *system identification*.

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TYPES OF SYSTEM

- lumped (described by odes)

or distributed (described by pdes)

- continuous time

or discrete time

- digital (discrete valued)

or analog (continuous valued)

- time invariant

or time varying

- linear

or non-linear

Of course some systems may have elements of each pair.

We consider mostly ;

lumped, continuous time, analog, time invariant, linear systems

TYPES OF MATHEMATICAL MODEL

(for analog systems)

(i) ODE

(ii) State Space

TOOLS FOR ANALYSIS AND DESIGN

Laplace transform

Fourier transform

State Space analysis

NB

Control relevant models are often quite simple compared to the true model.

But then so-called unmodelled dynamics must be kept in mind.

State Space

- State Space, Laplace, Fourier?

What is the state?

The *state* of a system represents the minimal amount of present information, about the history of the system, which is necessary to allow prediction of the effect of the past on the future.

Note that it is not unique.

All three system descriptions above are useful but the State Space (SS) description (or model) has special advantages:

- (i) handles higher order differential equation description easily
- (ii) handles time varying systems easily
- (iii) handles MIMO easily
- (iv) easy to program
- (v) captures initial conditions compactly (can keep track of hidden modes)
- (vi) handles noise components easily. SS is then a special case of Markov models.

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Modelling and Linearization: Water Buckets

Outflow Law

$$y(t) = q\sqrt{x(t)}$$

Conservation of mass

$$c\dot{x} = u(t) - y(t)$$

These give a nonlinear SS model immediately

$$\begin{aligned}\dot{x} &= -\frac{q}{c}\sqrt{x(t)} + \frac{1}{c}u(t) \\ y(t) &= q\sqrt{x(t)}\end{aligned}$$

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Linearization Continued

Linearize about a nominal inflow $\bar{u}$. In steady state the corresponding nominal outflow is $\bar{y} = \bar{u}$ with nominal depth $\bar{x} = \frac{\bar{u}^2}{q^2}$. Do Taylor series expansions and use the deviations $x_d = x - \bar{x}$ etc to get

$$\begin{aligned}\dot{x}_d &= -\frac{1}{rc}x_d(t) + \frac{1}{c}u(t) \\ y(t) &= \frac{1}{r}x_d(t)\end{aligned}$$

with $r = \frac{2\bar{u}}{q^2}$. The transfer function is

$$P(s) = \frac{\frac{1}{rc}}{s + \frac{1}{rc}}$$

With 2 buckets in series we get a larger SS model, with state $x = [x_1, x_2]$

$$\begin{aligned}\dot{x} &= \begin{pmatrix} -\frac{1}{r_1 c_1} & 0 \\ \frac{1}{r_1 c_2} & -\frac{1}{r_2 c_2} \end{pmatrix} x(t) + \begin{pmatrix} \frac{1}{c_1} \\ 0 \end{pmatrix} u(t) \\ y(t) &= \begin{pmatrix} 0 & \frac{1}{r_2} \end{pmatrix} x(t)\end{aligned}$$

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MODEL STRUCTURE - ODE

- ODE

$$y'^{(n)} + a_1 y'^{(n-1)} + \dots + a_n y = b_1 u'^{(n-1)} + \dots + b_n u$$

with initial conditions $y_0'^{(n-1)}, y_0'^{(n-1)}, \dots, y_0$

Laplace Transform of ODE

$$a(s)\overline{y}(s) = b(s)\overline{u}(s) + IC$$

$$a(s) = s^n + a_1 s^{n-1} + \dots + a_n$$

$$b(s) = b_1 s^{n-1} + \dots + b_n$$

MODEL STRUCTURE -STATE SPACE

- State Space

$$\begin{aligned}\dot{x} &= Ax + bu \\ y &= c^T x + du \\ x(0) &= x_0\end{aligned}$$

Laplace Transform of SS

$$\begin{aligned}s\bar{x}(s) &= A\bar{x}(s) + b\bar{u}(s) + x_0 \\ \Rightarrow \bar{x}(s) &= (sI - A)^{-1}x_0 + (sI - A)^{-1}b\bar{u}(s) \\ \Rightarrow \bar{y}(s) &= c^T(sI - A)^{-1}x_0 + c^T(sI - A)^{-1}b\bar{u}(s) \\ &= \text{initial condition} + \text{forced response}\end{aligned}$$

So transfer function is

$$P(s) = c^T(sI - A)^{-1}b$$

N.B. The observed variable may not be the only signal of interest from a control point of view. Control aim may be directed at other related signals.

RELATING ODE TO SS

- Connexion

What is the relation between

$$P(s) = \frac{b(s)}{a(s)} \text{ and } P(s) = c^T(sI - A)^{-1}b?$$

- Which comes first? ODE or SS?

and how to convert between them

- Solutions to ODE and SS models

How to generate output signal from initial conditions and inputs?

SOLVING ODE

Simplest method is via LT. By partial fraction expansion. For simplicity suppose $a(s)$ has distinct roots

$$\begin{aligned} P(s) &= \sum_1^n \frac{\chi_r}{s + \lambda_r} \\ &\leftrightarrow \sum_1^n \chi_r e^{\lambda_r t} \\ \Rightarrow P(s)\bar{u}(s) &\leftrightarrow \sum_1^n \int_0^t \chi_r e^{\lambda_r(t-t')} u(t') dt' \end{aligned}$$

SOLVING SS

Solution can be developed easily in either time domain or via LT.

- **Time Domain Solution**

$$\begin{aligned}\frac{d}{dt}(e^{-At}x) &= e^{-At}bu \\ \Rightarrow x(t) &= e^{At}x_0 + \int_0^t e^{A(t-t')}bu(t')dt' \\ \Rightarrow y(t) &= c^T x(t) \\ &= c^T e^{At}x_0 + \int_0^t c^T e^{A(t-t')}bu(t')dt'\end{aligned}$$

- **Laplace Transform**

$$\begin{aligned}(sI - A)^{-1} &\leftrightarrow e^{At} \\ \Rightarrow c^T(sI - A)^{-1}b &\leftrightarrow c^T e^{At}b \\ \Rightarrow c^T(sI - A)^{-1}b\bar{u}(s) &\leftrightarrow \int_0^t c^T e^{A(t-t')}bu(t')dt'\end{aligned}$$

ODE $\rightarrow$ SS

A given ODE can be represented by many different SS models or 'canonical' forms. In practice only a few of these are useful. Here we discuss one.

• Controller Form or Realization or Representation

$$\begin{aligned}\ddot{y} + a_1\dot{y} + a_2y &= b_1\dot{u} + b_2u \\ \leftrightarrow a(s)\bar{y}(s) &= b(s)\bar{u}(s)\end{aligned}$$

Introduce the partial state $\xi(s)$ satisfying

$$\begin{aligned}\ddot{\xi} + a_1\dot{\xi} + a_2\xi &= u \\ \leftrightarrow a(s)\bar{\xi}(s) &= \bar{u}(s)\end{aligned}$$

Then use this to construct the state as

$$x_1 = \dot{\xi}, x_2 = \xi$$

ODE $\rightarrow$ SS Continued

so that

$$\begin{aligned}\dot{x} &= A_c x + b_c u \\ A_c &= \begin{pmatrix} -a_1 & -a_2 \\ 1 & 0 \end{pmatrix}, \quad b_c = \begin{pmatrix} 1 \\ 0 \end{pmatrix}\end{aligned}$$

Then from the ODE-LT we have

$$\begin{aligned}\bar{y}(s) &= b(s)\bar{\xi}(s) \\ \Rightarrow \bar{y}(s) &= b_1 s \bar{\xi}(s) + b_2 \bar{\xi}(s) \\ \Rightarrow \bar{y}(s) &= b_1 \bar{x}_1(s) + b_2 \bar{x}_2(s) \\ \Rightarrow y(t) &= b_1 x_1(t) + b_2 x_2(t) \\ \Rightarrow y(t) &= c_c^T x(t) \\ c_c^T &= [b_1, b_2]\end{aligned}$$

For other canonical forms see Kailath.

SS $\rightarrow$ ODE

$$\begin{aligned}P(s) &= c^T (sI - A)^{-1} b \\&= \frac{c^T \operatorname{adj}(sI - A) b}{\det(sI - A)}\end{aligned}$$

$$\begin{aligned}a(s) &= \text{characteristic polynomial} \\&= \det(sI - A)\end{aligned}$$

$$\begin{aligned}b(s) &= \text{numerator polynomial} \\&= c^T \operatorname{adj}(sI - A) b\end{aligned}$$

$$\Rightarrow P(s) = \frac{b(s)}{a(s)}$$

Proper & Strictly Proper Transfer Functions

A proper transfer function has
 $\text{degree}(b) \leq \text{degree}(a)$.

A strictly proper transfer function has
 $\text{deg}(b) < \text{deg}(a)$.

In the example just given we converted a strictly proper TF into SS form, $\dot{x} = Ax + bu, y = cx$.

To handle a proper TF we use polynomial division to make it strictly proper e.g.

$$\begin{aligned}\frac{b(s)}{a(s)} &= \frac{b_0 s^2 + b_1 s + b_0}{a_0 s^2 + a_1 s + a_2} \\ &= \frac{b(s) - \frac{b_0}{a_0} a(s)}{a(s)} + \frac{b_0}{a_0} \\ &= \frac{\bar{b}_1 s + \bar{b}_2}{a_0 s^2 + a_1 s + a_2} + \frac{b_0}{a_0}, \bar{b}_1 = b_1 - \frac{b_0}{a_0} a_1, \bar{b}_2 = b_2 - \frac{b_0}{a_0} a_2 \\ &= \text{strictly proper TF} + \text{constant} \\ &= c(sI - A)^{-1}b + d, d = \frac{b_0}{a_0}.\end{aligned}$$

Monic and Low Frequency Transfer Function Forms

The monic form of the TF has the highest degree coefficient of the denominator $a(s)$ to be 1.

$$\text{e.g. } P(s) = \frac{4s+6}{2s^2+s+5}$$
$$\Rightarrow P_{\text{monic}}(s) = \frac{2s+3}{s^2+\frac{1}{2}s+2\frac{1}{2}}.$$

The low frequency form of the TF has the constant in the denominator $a(s)$ to be 1.

$$\text{e.g. } P(s) = \frac{4s+6}{2s^2+s+5}$$
$$\Rightarrow P_{\text{lowFreq}}(s) = \frac{\frac{4}{5}s+\frac{6}{5}}{\frac{2}{5}s^2+\frac{1}{5}s+1}.$$

The name comes from the fact that

$$\lim_{j\omega \rightarrow 0} P(j\omega) = \frac{1}{5}$$

= coefficient of the constant in the numerator.

The monic form is useful theoretically.

The low frequency form is usually much more informative for practical use. Since one can read off the higher order coefficients directly.

Notation

Generally we denote a rational transfer function by upper case letters.

e.g. $P(s), C(s), H(s), G(s)$.

Polynomials are denoted by lower case letters.

e.f. $p(s), q(s), a(s), b(s), n(s), d(s)$.

State Space notation uses;

A, B, C, D or F, G, H, J and K or L for gain matrices; also p, q for eigenvectors.

But for SISO SS systems typical symbols for vectors are;

c, b, k ; and n for the SS dimension.

Generally potential confusion e.g. $n(s)$ versus n is avoided by context.

STATE SPACE TRANSFORMATION

- SS Model

$$\dot{x} = Ax + bu, y = c^T x$$

Let T be an $n \times n$ full rank matrix and let $x_T = Tx$ be a transformed state

- SS model for x_T is as follows:

$$\begin{aligned}\dot{x}_T &= T\dot{x} \\ &= TAT^{-1}Tx + Tbu \\ &= A_T x_T + b_T u \\ \Rightarrow A_T &= TAT^{-1}, b_T = Tb \\ y &= c^T x \\ &= c_T^T T^{-1}Tx \\ &= c_T^T x_T \\ \Rightarrow c_T &= T^{-T}c\end{aligned}$$

So, A, b, c have been transformed to

$$A_T, b_T, c_T = TAT^{-1}, Tb, T^{-T}c$$

But we now check that the input/output description is unchanged.

STATE SPACE TRANSFORMATION

Continued

- Transfer function of transformed model is

$$\begin{aligned}P_T(s) &= c_T^T (sI - A_T)^{-1} b_T \\&= c^T T^{-1} (sI - TAT^{-1})^{-1} Tb \\&= c^T [T^{-1} (sI - TAT^{-1}) T]^{-1} b \\&= c^T (sI - A)^{-1} b \\&= P(s)\end{aligned}$$

NB. What happens to initial conditions?

- Controllability matrix

$$\begin{aligned}\mathcal{C}_T &= \mathcal{C}(A_T, b_T) = [b_T, A_T b_T, A_T^2 b_T, \dots] \\&= [Tb, TAT^{-1}Tb, TAT^{-1}TAT^{-1}Tb, \dots] \\&= [Tb, TAb, TA^2b, \dots] \\&= T[b, Ab, A^2b, \dots] \\&= T\mathcal{C} = T\mathcal{C}(A, b)\end{aligned}$$

$$\Rightarrow T = \mathcal{C}_T \mathcal{C}^{-1} \text{ useful formula}$$

MODAL DESCRIPTION OF SS & ODE MODELS

Not generic but provides powerful (generic) intuition about properties and behaviour.

- SS model

$$\begin{aligned}\dot{x} &= Ax + bu \\ y &= c^T x + du (d = 0)\end{aligned}$$

Distinct eigenvalues $\Rightarrow$

$$\begin{aligned}Ap_k &= \lambda_k p_k \\ q_k^T A &= \lambda_k q_k^T \\ I &= \sum_1^n p_k q_k^T\end{aligned}$$

or in matrix form

$$\begin{aligned}AP &= P\Lambda \\ Q^T A^T &= Q^T \Lambda \\ I &= PQ^T = Q^T P \\ \Rightarrow A &= P\Lambda Q^T\end{aligned}$$

MODAL DESCRIPTION Continued

Introduce the 'modes'

$$\xi_k = q_k^T x \leftrightarrow \xi = Q^T x.$$

Now consider how the SS model is transformed

$$\begin{aligned} A, b, c &\rightarrow Q^T A Q^{-T}, Q^T b, Q^{-T} c \\ &= Q^T A P, Q^T b, P c \\ &= \Lambda, \beta, \gamma \end{aligned}$$

The LT output is now

$$\begin{aligned} \bar{y}(s) &= \gamma^T (sI - \Lambda)^{-1} \beta \bar{u}(s) \\ &= \sum_1^n \frac{\gamma_k \beta_k}{s - \lambda_k} \bar{u}(s) \end{aligned}$$

This exhibits output as sum of responses from each mode. Same as partial fraction expansion of transfer function.

CONTROLLABILITY - THEOREM

The SS system

$$\dot{x} = Ax + bu, y = c^T x$$

is controllable if given any initial state x_0 there is an input signal that drives the state to 0 in finite time.

The SS system is completely controllable if the state can be driven to any position x_1 in finite time.

In continuous time 'controllable' $\equiv$ 'completely controllable'.

In discrete time, they differ [reachability].e.g. $x_{k+1} = 0$ is controllable, but not completely controllable.

Theorem. The SS system (1) is completely controllable iff the controllability matrix $\mathcal{C}(A, b) = [b, Ab, \dots, A^{n-1}b]$ has full rank.

CONTROLLABILITY - PROOF

Proof. We construct $u(t)$.

The solution at time t is

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-t')}bu(t')dt'$$

Choose $u(t') = b^T e^{A^T(T-t')}M_T^{-1}(x_1 - e^{AT}x_0)$

with M_T to be chosen. Then

$$\begin{aligned} x(T) &= e^{AT}x_0 \\ &+ \int_0^T e^{A(T-t')}bb^T e^{A^T(T-t')}dt' [M_T^{-1}(x_1 - e^{AT}x_0)] \end{aligned}$$

if we set [change variables to $v = T - t'$]

$$M_T = \int_0^T e^{Av}bb^T e^{A^Tv}dv$$

Then we find $x(T) = x_1$, as required.

CONTROLLABILITY PROOF Continued

Now we need to show M_T has full rank. Suppose not: Then there exists a vector a with

$$\begin{aligned} a^T M_T a &= 0 \Rightarrow \int_0^T (a^T e^{Av} b)^2 dv = 0 \\ \Rightarrow a^T e^{Av} b &= 0, 0 \leq v \leq T \end{aligned}$$

Differentiate repeatedly to find

$$\begin{aligned} a^T e^{Av} b &= 0, a^T A e^{Av} b = 0, a^T A^2 e^{Av} b = \\ 0, \dots, a^T A^{n-1} e^{Av} b &= 0 \end{aligned}$$

set $v = 0$ and collect these together

$$\Rightarrow a^T [b, Ab, \dots, A^{n-1}b] = a^T \mathcal{C} = 0 \text{ i.e. } \mathcal{C} \text{ singular}$$

So $\mathcal{C}$ full rank $\Rightarrow M_T$ full rank

For the reverse case. If M_T full rank but not $\mathcal{C}$ then there is a vector a with $a^T \mathcal{C} = 0$

$$\Rightarrow a^T A^m b = 0, 0 \leq m \leq n - 1$$

Cayley-Hamilton Theorem now gives

$$\begin{aligned} a^T A^k b &= 0, \text{ all } k \Rightarrow a^T e^{At} b = 0, \text{ all } t \\ \Rightarrow a^T M_T a &= \int_0^T (a^T e^{Au} b)^2 du = 0 \end{aligned}$$

So M_T is singular.

NB. M_T is called the controllability Grammian.

CONTROLLABILITY - Example

Two buckets in series

We saw earlier

$$A, b = \begin{pmatrix} -\tau_1 & 0 \\ +\tau_{12} & -\tau_2 \end{pmatrix}, \begin{pmatrix} b_1 \\ 0 \end{pmatrix}$$

$$\Rightarrow Ab = \begin{pmatrix} -\tau_1 b_1 \\ \tau_{12} b_1 \end{pmatrix}$$

$$\Rightarrow \mathcal{C}(A, b) = \begin{pmatrix} b_1 & -\tau_1 b_1 \\ 0 & -\tau_{12} b_1 \end{pmatrix}$$

$$\Rightarrow |\mathcal{C}(A, b)| = -\tau_{12} b_1^2 \neq 0$$

So $\mathcal{C}(A, b)$ has full rank

so SS system is controllable.

OBSERVABILITY - THEOREM

The SS system $\dot{x} = Ax + bu, y = c^T x$

is observable at time 0 if the initial state can be uniquely determined from the input, output record over a finite time.

The SS system is reconstructable if the state at any time can be uniquely recovered.

In continuous time observability $\equiv$ reconstructability.

In discrete time they differ.

Theorem. SS system is observable iff the observability matrix

$$\mathcal{O}(c, A) = \begin{bmatrix} c^T \\ c^T A \\ : \\ : \\ c^T A^{n-1} \end{bmatrix}$$

has full rank.

OBSERVABILITY - PROOF

Integrate SS equation to find, with $u(t) = 0$

$$\int_0^T e^{A^T t} c y(t) dt = \int_0^T e^{A^T t} c c^T e^{A t} x_0 dt = G_0 x_0$$

So we can recover x_0 iff G_0 is of full rank. If G_0 is singular there is a position $x_a \neq 0$ with

$$\begin{aligned} G_0 x_a = 0 &\Rightarrow x_a^T G_0 x_a = 0 \Rightarrow \int_0^T (c^T e^{A t} x_a)^2 dt = 0 \\ &\Rightarrow c^T e^{A t} x_a = 0, \quad 0 \leq t \leq T \end{aligned}$$

Repeated differentiation gives

$$\begin{bmatrix} c^T \\ c^T A \\ \vdots \\ c^T A^{n-1} \end{bmatrix} e^{A t} x_a = 0$$

Setting $t=0$ gives that $\mathcal{O}(c, A)$ is singular.

OBSERVABILITY PROOF Continued

For the reverse if there is ξ with

$$\mathcal{O}\xi = 0 \Rightarrow c^T A^m \xi = 0, \quad 0 \leq m \leq n-1$$

Then Cayley Hamilton theorem gives

$$c^T e^{At} \xi = 0 \Rightarrow \xi^T G_0 \xi = \int_0^T (c^T e^{At} \xi)^2 dt = 0$$

So, G_0 is singular

So, G_0 full rank iff $\mathcal{O}(c, A)$ full rank.

MODAL VIEW OF CONTROLLABILITY, OBSERVABILITY

For the SS model

$$\dot{x} = Ax + bu, \quad y = c^T x$$

Suppose A has distinct eigenvalues, then the eigenvalue decomposition is

$$A = P\Lambda Q^T, \quad Q^T P = I = PQ$$

and the transformed SS model with state

$$\xi = Q^T x \text{ is}$$

$$\begin{aligned}\dot{\xi} &= Q^T \dot{x} = Q^T P\Lambda Q^T x + Q^T bu \\ &= \Lambda \xi + \beta u, \quad [\beta = Q^T b]\end{aligned}$$

$$\Rightarrow \dot{\xi}_r = \lambda_r \xi_r + \beta_r u, \quad r = 1, \dots, n$$

i.e. each state equation involves only its own state, these are the 'independent' modes. The output can be expressed as

$$y = c^T P Q^T x = c^T P \xi = \gamma^T \xi \quad [\gamma = P^T c]$$

$$\Rightarrow y(t) = \sum_1^n \gamma_r \xi_r(t)$$

=weighted sum of modes

MODAL VIEW Continued

Controllability

$$\begin{aligned}\mathcal{C}(\Lambda, \beta) &= (\beta, \Lambda\beta, \dots, \Lambda^{n-1}\beta) \\ &= \begin{pmatrix} \beta_1 & \lambda_1\beta_1 & : & \lambda_1^{n-1}\beta_1 \\ \beta_2 & \lambda_2\beta_2 & : & \lambda_2^{n-1}\beta_2 \\ : & : & : & : \\ \beta_n & \lambda_n\beta_n & : & \lambda_n^{n-1}\beta_n \end{pmatrix}\end{aligned}$$

Now controllability requires $\mathcal{C}(\Lambda, \beta)$ has full rank
i.e. no linear dependencies among columns, rows
of $\mathcal{C}(\Lambda, \beta)$

This clearly occurs iff

$\lambda_k \neq \lambda_\ell$ (rules out, column-wise dependence)

$\beta_k \neq 0$ (rules out, row-wise dependence)

If $\beta_r = 0$, then the input does not effect the r th
mode: so $\dot{\xi}_r = \lambda_r \xi_r + \beta_r u = \lambda_r \xi_r$

i.e. the r th mode cannot be manipulated by the
control signal.

This gives very explicit meaning to lack of
controllability.

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MODAL OBSERVABILITY

The observability matrix is

$$\mathcal{O}(\gamma, \Lambda) = \begin{bmatrix} \gamma^T \\ \gamma^T \Lambda \\ \vdots \\ \gamma^T \Lambda^{n-1} \end{bmatrix} = \begin{bmatrix} \gamma_1 & \gamma_2 & \vdots & \gamma_n \\ \lambda_1 \gamma_1 & \lambda_2 \gamma_2 & \vdots & \lambda_n \gamma_n \\ \vdots & \vdots & \vdots & \vdots \\ \lambda_1^{n-1} \gamma_1 & \lambda_2^{n-1} \gamma_2 & \vdots & \lambda_n^{n-1} \gamma_n \end{bmatrix}$$

Again full rank for $\mathcal{O}(\gamma, \Lambda)$ requires no row or column dependencies. This entails

$\gamma_i \neq 0$ (rules out, column dependencies)

$\lambda_k \neq \lambda_\ell$ (rules out, row dependencies)

The meaning is clear. Since

$$y(t) = \gamma^T \xi(t) = \sum_1^n \gamma_r \xi_r(t)$$

Then $\gamma_r = 0$ means the r th mode does not appear in $y(t)$ and so its initial condition cannot be got from $y(t)$.

MODAL VIEW OF POLE - ZERO CANCELLATION

For a proper TF $P(s) = \frac{b(s)}{a(s)}$;

A TF *zero* is a value s_0 with $b(s_0) = 0$.

A TF *pole* is a value s_p with $a(s_p) = 0$.

If a TF $P(s)$ has a pole-zero cancellation then its order (i.e. the order of $a(s)$) drops by one.

Insight from a modal view is available from the TF (transfer function) expression

$$\bar{y}(s) = \sum_1^n \gamma_k \beta_k \frac{\bar{u}(s)}{s - \lambda_k} = P(s) \bar{u}(s)$$

If $\beta_k = 0$ or $\gamma_k = 0$ we get a mode missing. This means the partial fraction expansion of $P(s)$ has only $n - 1$ terms. This means there must be a pole-zero cancellation in $P(s)$.

HIDDEN MODES

Now we include initial conditions in our computations. Using the modal SS model the Laplace transform is

$$\begin{aligned}\bar{y}(s) &= \gamma^T (sI - \Lambda)^{-1} \xi_0 + \gamma^T (sI - \Lambda)^{-1} \beta \bar{u}(s) \\ &= \sum_1^n \frac{\gamma_k \xi_{0k}}{s - \lambda_k} + \sum_1^n \frac{\gamma_k \beta_k}{s - \lambda_k} \bar{u}(s)\end{aligned}$$

Now if there is an uncontrollable mode so that $\beta_r = 0$ we get a pole-zero cancellation and the TF does not show the hidden mode (hence the name!) However the hidden mode remains in the IC (except in the unlikely event $\xi_{0k} = 0$) and can cause problems.

With pz cancellation due to lack of observability the mode does completely drop out of both terms. It may of course affect other combinations of the state variables.

MODAL EVD EXAMPLE

For the SS model

$$A = \begin{bmatrix} -1 & 0 \\ 1 & -2 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, c = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Find the modal decomposition and investigate controllability, observability, transfer function, stability.

Eigenvalues solve:

$$0 = |\lambda I - A| = \begin{vmatrix} \lambda + 1 & 0 \\ -1 & \lambda + 2 \end{vmatrix} = (\lambda + 1)(\lambda + 2)$$

$\Rightarrow \lambda_1, \lambda_2 = -1, -2 \Rightarrow$ real roots (both stable).

Eigenvectors satisfy

$$Ap^{(i)} = \lambda_i p^{(i)}, i = 1, 2$$

For λ_1 we find

$$\Rightarrow \begin{pmatrix} -1 & 0 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} p_1^{(1)} \\ p_2^{(1)} \end{pmatrix} = -1 \begin{pmatrix} p_1^{(1)} \\ p_2^{(1)} \end{pmatrix}$$

$$\Rightarrow -p_1^{(1)} = -p_1^{(1)}, p_1^{(1)} - 2p_2^{(1)} = -p_2^{(1)}$$

$$\Rightarrow p_1^{(1)} = p_2^{(1)}$$

EXAMPLE - Continued

$$\Rightarrow p^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ (need not be unit length)}$$

For λ_2 we find

$$\begin{pmatrix} -1 & 0 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} p_1^{(2)} \\ p_2^{(2)} \end{pmatrix} = -2 \begin{pmatrix} p_1^{(2)} \\ p_2^{(2)} \end{pmatrix}$$

$$\Rightarrow -p_1^{(2)} = -2p_1^{(2)} \Rightarrow p_1^{(2)} = 0$$

$$p_1^{(2)} - 2p_2^{(2)} = -2p_2^{(2)} \Rightarrow p^{(2)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\Rightarrow P = [p^{(1)}, p^{(2)}] = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

$$\Rightarrow Q^T = P^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} q_1^T \\ q_2^T \end{pmatrix}$$

EXAMPLE Continued

Controllability matrix is

$$\begin{aligned}\mathcal{C}(A, b) &= [b, Ab] \\ &= \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{pmatrix} -1 & 0 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right) \\ &= \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}\end{aligned}$$

which is clearly singular so SS model is not controllable.

Modal view : Control coefficients are

$$\beta_1 = q_1^T b = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1$$

$$\beta_2 = q_2^T b = \begin{pmatrix} -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0$$

So second mode is not controllable. It is stable so SS model is stabilizable.

EXAMPLE - Continued

Observability

$$\begin{aligned}\mathcal{O}(c, A) &= \begin{pmatrix} c^T \\ c^T A \end{pmatrix} \\ &= \begin{bmatrix} 1 & 1 \\ \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 1 & -2 \end{pmatrix} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & -2 \end{bmatrix}\end{aligned}$$

which has full rank (e.g. its $\det \neq 0$)

So SS model is observable.

Modal View

$$\begin{aligned}\gamma &= P^T c = \begin{pmatrix} 1 & +1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} \gamma_1 \\ \gamma_2 \end{pmatrix}\end{aligned}$$

So both $\gamma_1, \gamma_2, \neq 0$ which confirms observability.

EXAMPLE - Continued

Transfer Function can now be computed from

$$P(s) = \sum_1^2 \frac{\gamma_r \beta_r}{s - \lambda_r} = \frac{2 \times 1}{s+1} + \frac{1 \times 0}{s+2} = \frac{2}{s+1}$$

So as expected due to lack of controllability there is a pole zero cancellation. Alternatively, from a SS view

$$\begin{aligned} P(s) &= c^T (sI - A)^{-1} b \\ &= \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} s+1 & 0 \\ -1 & s+2 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \frac{1}{(s+1)(s+2)} \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} s+1 & 0 \\ 1 & s+2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \frac{1}{(s+1)(s+2)} \begin{pmatrix} s+3 & s+1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \frac{1}{(s+1)(s+2)} (2s+4) = \frac{2}{(s+1)} \frac{(s+2)}{(s+2)} = \frac{2}{s+1} \end{aligned}$$

Here we see the pole zero cancellation precisely.

N.B. Write out initial condition, Laplace transform showing the hidden mode.

APPENDIX 2A: FINAL VALUE THEOREM

If all the poles of $s\bar{y}(s)$ are in the left half plane
then

$$y(\infty) = y_\infty = \lim_{s \rightarrow 0} s\bar{y}(s)$$

Proof: For simplicity suppose a partial fraction
expansion holds with distinct roots

$$s\bar{y}(s) = c + \sum_1^n \frac{\theta_k}{s - \lambda_k}$$

Then on the one hand

$$\lim_{s \rightarrow 0} s\bar{y}(s) = c - \sum_1^n \frac{\theta_k}{\lambda_k}$$

APPENDIX 2A - Continued

On the other hand

$$\begin{aligned}\bar{y}(s) &= \frac{c}{s} + \sum_1^n \frac{\theta_k}{s(s - \lambda_k)} \\ &= \frac{c}{s} + \sum_1^n \frac{\theta_k}{\lambda_k} \left[\frac{1}{s - \lambda_k} - \frac{1}{s} \right] \\ &= \frac{1}{s} \left(c - \sum_1^n \frac{\theta_k}{\lambda_k} \right) + \sum_1^n \frac{\theta_k}{\lambda_k} \frac{1}{s - \lambda_k}\end{aligned}$$

Now take inverse LT to find

$$y(t) = c - \sum_1^n \frac{\theta_k}{\lambda_k} + \sum_1^n \frac{\theta_k}{\lambda_k} e^{\lambda_k t}$$

And since by assumption $Re(\lambda_k) < 0$ we find

$$\lim_{t \rightarrow \infty} = c - \sum_1^n \frac{\theta_k}{\lambda_k} + 0$$

and the result follows.

APPENDIX 2B: PBH TESTS

The Popov-Belevitch-Hautus (PBH) tests generalise the idea of modal view of controllability and observability. They are often quite powerful in practice.

PBH Controllability.

A, b is controllable iff there does not exist non-zero q with

$$q^T A = \lambda q^T, \text{ and } q^T b = 0$$

i.e. there is no left eigenvector of A orthogonal to b .

PBH Observability

(c, A) is observable iff there is no non-zero vector p with

$$Ap = \lambda p, \text{ } p^T c = 0$$

i.e no right eigenvector of A is orthogonal to c .

APPENDIX 2B Continued

- Proof for Controllability: This is intuitively clear from the modal view. If such a q exists then note that

$$\begin{aligned} q^T A^r &= \lambda q^T A^{(r-1)} = \dots = \lambda^r q^T \\ \Rightarrow q^T \mathcal{C}(A, b) &= (q^T b, q^T A b, \dots, q^T A^{(n-1)} b) \\ &= (q^T b, \lambda q^T b, \dots, \lambda^{(n-1)} q^T b) \\ &= q^T b (1, \lambda, \dots, \lambda^{(n-1)}) \\ &= 0^T \end{aligned}$$

So $\mathcal{C}(A, b)$ is singular.

Proof of converse is more complex, uses the 'decomposition' theorem and will be given later.

- Proof for Observability: Similar to controllability - left as an exercise.